

VOSTOCHNAYA, Russian Federation

WMO: 246790

Lat: 63.217N

Lon: 139.600E

Elev: 1288

StdP: 86.78

Time Zone: 9.00 (E09)

Period: 86-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-45.3	-43.2	-47.3	0.0	-45.1	-45.3	0.0	-43.1	9.5	-31.7	8.7	-32.3	1.0	210	N/A

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.8	21.5	12.3	19.6	11.2	17.8	10.5	13.4	18.7	12.4	17.4	11.5	16.1	3.0	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
11.4	9.8	14.3	10.4	9.2	13.5	9.3	8.5	13.0	41.8	18.7	38.9	17.4	36.3	16.0	17.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 9.6	8.2	7.0	DB	-46.7	25.4	2.5	2.1	-48.5	26.9	-49.9	28.1	-51.3	29.3	-53.1	30.8	(4)
(5)			WB	-46.7	15.1	2.4	1.2	-48.4	16.0	-49.8	16.7	-51.1	17.4	-52.9	18.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-12.5	-34.0	-31.4	-23.4	-13.3	-1.7	8.3	11.2	8.6	-0.1	-14.9	-27.5	-32.9	(6)	
	DBStd	17.33	5.33	5.36	5.80	4.69	4.90	3.94	3.58	3.51	4.76	6.07	6.03	5.45	(7)	
	HDD10.0	8325	1363	1161	1034	698	364	77	29	67	302	773	1126	1332	(8)	
	HDD18.3	11252	1622	1394	1293	948	622	302	222	301	552	1032	1376	1590	(9)	
	CDD10.0	116	0	0	0	0	0	25	66	24	1	0	0	0	(10)	
	CDD18.3	1	0	0	0	0	0	0	1	0	0	0	0	0	(11)	
	CDH23.3	16	0	0	0	0	0	5	11	0	0	0	0	0	(12)	
	CDH26.7	0	0	0	0	0	0	0	0	0	0	0	0	0	(13)	
(14) Wind	WSAvg	2.6	2.0	2.2	2.6	3.0	3.1	3.0	2.7	2.5	2.6	2.7	2.5	1.9	(14)	
(15) Precipitation	PrecAvg	299	3	2	3	10	20	54	66	54	41	15	7	5	(15)	
	PrecMax	411	9	7	8	25	59	108	208	109	78	32	18	12	(16)	
	PrecMin	186	0	0	0	1	9	8	20	19	14	1	0	1	(17)	
	PrecStd	54	2	2	2	6	11	27	42	23	19	7	5	3	(18)	
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-18.0	-15.5	-7.5	1.4	13.2	23.8	24.9	21.8	14.1	-0.4	-8.5	-17.6	(19)	
		MCWB	-18.5	-15.7	-9.1	-2.1	5.8	12.5	13.5	12.4	7.0	-1.8	-9.3	-18.2	(20)	
	2%	DB	-21.7	-18.8	-9.5	-1.2	10.6	20.7	22.4	19.5	11.8	-2.3	-13.5	-20.4	(21)	
		MCWB	-21.9	-19.1	-10.8	-3.7	4.2	11.1	13.1	11.2	6.0	-3.7	-14.0	-20.7	(22)	
	5%	DB	-24.1	-21.7	-11.8	-3.4	8.7	18.8	20.2	17.7	9.7	-3.9	-16.3	-22.6	(23)	
		MCWB	-24.3	-22.0	-12.9	-5.7	3.1	9.7	12.0	10.6	4.9	-4.9	-16.7	-22.9	(24)	
	10%	DB	-25.9	-24.0	-13.8	-5.2	6.2	16.7	18.4	15.6	7.5	-5.8	-18.9	-24.6	(25)	
		MCWB	-26.1	-24.4	-14.8	-7.1	1.9	8.9	11.3	9.7	3.8	-6.8	-19.3	-24.9	(26)	
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-18.7	-15.7	-8.8	-1.6	6.4	13.0	15.8	13.3	8.2	-1.5	-9.3	-18.2	(27)	
		MCDB	-18.4	-15.5	-7.7	1.1	12.5	21.6	22.4	18.6	11.8	-0.5	-8.6	-17.8	(28)	
	2%	WB	-22.1	-19.1	-10.8	-3.8	4.6	11.5	14.0	12.2	6.6	-3.6	-14.0	-20.7	(29)	
		MCDB	-21.9	-18.9	-9.5	-1.5	9.6	18.4	19.2	17.2	10.6	-2.6	-13.6	-20.5	(30)	
	5%	WB	-24.3	-22.0	-12.9	-5.6	3.4	10.7	13.2	11.4	5.3	-5.0	-16.7	-22.8	(31)	
		MCDB	-24.2	-21.7	-11.8	-3.6	7.9	16.8	18.1	15.9	9.1	-4.0	-16.4	-22.6	(32)	
	10%	WB	-26.1	-24.3	-14.8	-6.9	2.3	9.8	12.3	10.7	4.0	-6.8	-19.2	-24.8	(33)	
		MCDB	-25.8	-24.0	-13.8	-5.2	6.1	15.3	16.8	14.6	7.1	-5.9	-18.9	-24.6	(34)	
(35) Mean Daily Temperature Range	MDBR		6.4	8.7	11.6	12.2	9.4	10.7	9.8	10.0	8.5	8.1	6.9	5.8	(35)	
	5% DB	MCDBR	7.5	9.7	12.4	13.0	12.7	15.9	14.1	14.6	12.1	7.3	9.3	7.4	(36)	
		MCWBR	7.4	9.4	11.1	10.5	7.6	7.8	6.8	7.2	6.3	6.3	8.9	7.2	(37)	
	5% WB	MCDBR	7.7	9.7	12.2	12.0	11.1	13.5	11.7	10.9	10.7	7.2	9.2	7.4	(38)	
		MCWBR	7.5	9.4	10.8	9.7	6.8	7.2	6.3	6.1	6.7	6.1	8.8	7.2	(39)	
(40) Clear-Sky Solar Irradiance	taub		0.159	0.217	0.265	0.322	0.360	0.346	0.391	0.354	0.280	0.229	0.167	0.145	(40)	
	taud		1.995	2.169	2.108	1.983	2.039	2.387	2.270	2.360	2.440	2.218	1.959	1.908	(41)	
	Ebn at Noon		539	759	834	845	843	874	814	818	831	722	484	308	(42)	
	Edh at Noon		29	59	97	139	144	105	115	96	71	55	29	14	(43)	
(44) All-Sky Solar Radiation	RadAvg		0.27	1.24	3.13	5.26	5.73	5.31	4.91	3.77	2.30	1.34	0.46	0.11	(44)	
	RadStd		0.02	0.05	0.11	0.20	0.32	0.45	0.45	0.30	0.23	0.10	0.04	0.01	(45)	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page